

Quorum Configuration Changes and Liveness Margin in the Stellar Network, 2019–2026

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Summary

We analysed 1.2 million network scans of the Stellar public network from May 2019 to October 2026, checked every period of an hour or more in which the liveness margin was reduced against a full network snapshot, and pulled daily availability for every organization that was in the top tier. Safety margins have been stable since February 2020: quorum intersection held in every scan. Liveness margins were not stable. Four times, one operator changed its quorum set while no other operator changed theirs, and the fault tolerance of the whole network dropped, twice from two organization failures to one, for between five hours and 7.5 days. Separately, three of the four organizations that left the top tier over the period (Keybase, Wirex and SatoshiPay; not Whalestack) showed months of degraded validator availability first, and SatoshiPay accounts for ten of the sixteen verified organization outages of an hour or more. A simple availability rule scored across all top tier organizations warns of 62% of outages and removals at least a week ahead, with a median lead of about six months, and 62% of its warnings are followed by an outage or removal between one week and one year later; without SatoshiPay, whose chronic degradation supplies most of the hits, the precision falls to 35%. Availability alone is therefore a weak early warning signal; the configuration side is where a check would have made a difference.

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1. Data and method

All data come from the public API of OBSRVR Radar, formerly Stellarbeat [1].

- **Network statistics.** 1,214,403 scans from 31 May 2019 to 1 October 2026, one every three to five minutes. Each scan records the top tier size, the minimal blocking set and minimal splitting set [2] at node and organization level, and a second blocking set figure that discounts validators currently failing.
- **Network snapshots.** Full network state at 159 instants: the start of each of the 52 stable configuration states, and one to four instants inside every event. Each snapshot gives every validator’s quorum set and validating status.
- **Daily availability.** Per validator and per organization daily availability for all 14 organizations that were in the top tier for at least a month, over the whole period.

Throughout, the *liveness margin* is the minimal blocking set in organizations, the number of organizations whose failure would halt the network, and the fault tolerance is one less. The *nominal* margin is computed from the configurations Radar sees; the *effective* margin discounts validators that are not validating at the time. Both are computed over the top tier, the transitive quorum set.

Events were detected in two ways. A *configuration event* is a change in the nominal margins that persists for at least five scans and that a snapshot diff attributes to a changed quorum set. A *degradation event* is a period in which the effective margin is below the nominal margin; periods separated by less than 30 minutes are merged, so a period of an hour may contain less than an hour of reduced margin. Every event was checked against snapshots: an outage is counted only if the validators of a named organization were not validating while the rest of the top tier was. Scans in which more than half of the top tier appears down at once are labelled crawler outages, and ledger close times from StellarExpert [10] were used as an independent check of whether the network itself was affected.

One property of the data matters for interpretation. Radar’s nominal margins respond not only to quorum set changes but also to reachability: a validator the crawler cannot reach at all drops out of the nominal computation. Two of the six episodes flagged as weak intermediate states by the scan (February 2022 and July

2022) were of this kind, with no quorum set change in any top tier validator, and the same effect delays the start of a degradation event when a failed validator is also unreachable. This is why the snapshot diff is part of the definition above.

For every configuration event we recomputed the minimal blocking set ourselves by exhaustive enumeration over the top tier. Our figures match those recorded by Radar in every case, before and after the change.

2. Finding 1: safety margins have been stable since February 2020

Quorum intersection held in all 1,214,403 scans. From February 2020 to August 2026 the minimal splitting set of the top tier stayed at three organizations (it was four in late 2019, when the top tier was smaller); it rose to four after the expansion to ten tier 1 organizations on 27 and 28 August 2026. These are top tier figures. Network wide, counting validators outside the top tier that can be cut off from the core more cheaply than the core can be split, Radar’s current analysis reports two organizations [9].

Top tier membership changed nine times: Blockdaemon joined (October 2019), Wirex joined (January 2020), Keybase left and Public Node joined (March 2022), Franklin Templeton joined (October 2022), Wirex left (December 2022), Whalestack, formerly COINQVEST, left and Creit Technologies joined (January 2025), LOBSTR retired two of its five validators (April 2025), SatoshiPay left and OBSRVR joined (15 July 2026), and MoneyGram, Range and Figure joined (27 August 2026). None of these changes passed through an intermediate state with a lower organization level margin than both its starting and its final configuration. They were, however, staggered: in the August 2026 expansion the three new organizations appeared in the transitive quorum set on 27 August at 09:23 UTC, but the blocking set rose from three organizations to four only on 28 August at 08:26, once the last operators had raised their thresholds. For about 23 hours the network had ten organizations and still tolerated the failure of only two.

The history therefore does not support the premise of gradual erosion of safety. The events worth studying are on the liveness side.

3. Finding 2: a single operator’s change lowered the fault tolerance of the whole network

Four times, one operator changed its quorum set while no other operator changed theirs, and the fault tolerance of the whole network dropped (Table 1). Two of the changes lowered the margin at organization level, from tolerating two organization failures to tolerating one; two lowered it at node level only. Three were made in normal operation, with every organization validating; SDF’s began while its validators were recovering from the halt of 6 April 2021. In three cases the operator removed one organization from its list; in the fourth, two LOBSTR validators wrapped Blockdaemon together with two organizations outside the top tier into an inner set that required all three of them, which pulled seven outside validators into the transitive quorum set for six hours. All four changes were later reverted and the margins returned to their previous values.

Period (UTC)	Operator (validators)	Change	Quorum set	Blocking set	Duration
22 to 29 May 2020	Blockdaemon (3)	removed SatoshiPay	5 of 7 to 5 of 6	3/6 to 2/4	7.5 days
6 to 7 April 2021	SDF (3)	removed LOBSTR	5 of 7 to 5 of 6	3/6 to 2/4	1.3 days
13 April 2021	LOBSTR (2 of 5)	3 of 3 inner set	5 of 7, nested	3/6 to 3/5	5.4 hours
3 to 8 October 2024	Franklin Templeton (1 of 3)	removed LOBSTR	5 of 7 to 5 of 6	3/6 to 3/5	4.6 days

Table 1: Single operator changes that lowered the network wide liveness margin (blocking set in organizations/nodes).

Mechanism. `stellar-core` generates quorum sets automatically and gives a group of n organizations the threshold $n - \lfloor (n - 1)/3 \rfloor$, which assumes Byzantine failure [3, 4]. For seven organizations the threshold is five, and for six it is still five. An operator who deletes one organization from its list therefore ends up requiring five of the remaining six. Two failures among those six now block that operator. Once it is blocked, every other organization has lost three of its seven and is blocked as well.

Counterfactual. For each case we took the configuration before the change and replaced only the quorum sets of the validators that changed. The recomputed minimal blocking set equals the value observed after the

change in all four cases. The change alone is sufficient to explain the drop, and a check run on the proposed quorum set before applying it would have shown the effect.

Operational consequences. The lowered margin was not theoretical. During the 7.5 days of Blockdaemon’s change the effective margin fell to one organization, Radar’s alert level, in four separate periods of 15 minutes or more, each caused by a single organization’s validators faltering; before and after the change the same reading required two organizations to be down at once. On 4 October 2024, with FT SCV 2 on its reduced quorum set, the other Franklin Templeton validator FT SCV 1 stopped for about two hours and the effective organization margin fell to two; under the previous quorum set the same failure leaves it at three. The change turned one validator’s failure into the loss of an organization of margin.

Operator intent is unknown, and the changes were defensible from one seat. In two of the four cases the removed organization had just been unstable: SatoshiPay showed degraded availability on 22 of the 33 days before Blockdaemon dropped it, and LOBSTR had been down together with SDF on 6 April 2021. SDF’s change began during that incident. Radar records SDF’s and LOBSTR’s validators as not validating from 14:34 to 18:27 UTC; SDF’s statement reports its own validators halted from about 08:00 to 18:28 UTC and restored by bringing new nodes online, and does not mention LOBSTR [6]. The network margin dropped at 17:22 UTC on 6 April as SDF’s validators came back; by 11:57 UTC on 7 April they ran a quorum set of five of the six organizations other than LOBSTR, and the earlier configuration returned at 23:38 UTC that day. At least three operators (SDF, LOBSTR, SatoshiPay) changed quorum sets unilaterally in the fortnight after the incident. Removing a peer that has just failed is a reasonable decision for one operator; what it costs the network is invisible from there.

Existing safeguards. The stellar-core configuration guide warns that removing a validator may have unintended consequences for other participants and advises coordination [3]. stellar-core can check quorum intersection for the running network and offline for a proposed configuration, but that is a safety check [5, 4]. Stellarbeat has offered, since at least February 2021, a simulation mode in which a user can edit one node’s quorum set by hand and rerun the network analysis [7]; fbas_analyzer [2] and python-fbas [8] can analyse any configuration given to them. What does not exist is a check that runs on a proposed change inside an operator’s workflow and reports its effect on the liveness margin of the whole network before the change is applied.

4. Finding 3: outages were preceded by months of visible degradation, and most degradation was not followed by an outage

The scan contains 32 periods of an hour or more in which failing validators reduced the effective margin. Snapshots verify 16 of them as outages of a named organization (Table 2); 12 are short dips that the sampled snapshot did not catch at the sampled instant; three fall on 14 September 2023, when Radar’s crawler lost most of the top tier for hours while the network kept closing ledgers, at 8.2 seconds per ledger between 10:27 and 10:40 UTC against 5.9 seconds at the same hour the day before, so the network was disturbed but not stopped, and we exclude these periods because they cannot be attributed to an organization; one is an attribution artifact in which Radar grouped a LOBSTR validator under a separate home domain. SatoshiPay accounts for ten of the sixteen verified outages, spread over 2020, 2021, 2024 and 2025.

Organization	Verified outages within periods of an hour or more	Longest	Margin during the longest
SDF	23 April 2020; 10 May 2020	7.7 hours	2 organizations, briefly 1
SDF and LOBSTR	6 April 2021, 8 of 23 validators down together	3.9 hours	1 organization
Wirex	13 July, 26 July and 13 December 2022	1.2 days	2 organizations, briefly 1
SatoshiPay	31 May 2020; 1 September (twice) and 13 October 2021; 14 February 2024; 18, 25 (twice) and 26 February 2025; 22 August 2025	3.1 days	2 organizations

Table 2: Organization outages verified against snapshots. Wirex left the top tier on 14 December 2022; SatoshiPay was removed from quorum sets on 15 July 2026.

Radar’s network alert for organization level liveness fires when the effective margin reaches one organization [1]. Among the 32 periods, only the incident of 6 April 2021 held at that level for hours; five others touched it for minutes. Across all scans the effective margin was at one organization or below in 115 short runs, most of them single scans in 2019 when the nominal margin was two.

Degradation before exit. Three of the four organizations that left the top tier degraded first. Keybase had one validator at or near zero availability for days at a time from October 2021, and continuously from late February 2022, and was removed from quorum sets between 8 and 22 March 2022, as Public Node took its place. Wirex showed intermittent validator outages from 10 July 2022, a two day organization outage in late July, and left on 14 December 2022. SatoshiPay showed degraded daily availability on 36 days between January and March 2025, a three day outage from 22 August 2025, renewed degradation from December 2025, and was removed on 15 July 2026; its validators stopped altogether on 22 July 2026. Whalestack, which left in January 2025 as Creit Technologies took its place, showed only minor signs beforehand.

Scoring across all organizations. Lead times measured only on organizations already known to have failed say nothing about false alarms. We therefore scored candidate rules over all 14 organizations and all 18,592 organization days in the top tier. Rule S3 fires on a day on which any validator of the organization validated for 10% of the day or less, or the organization as a whole was available for less than 90% of it; rule S2 fires when the organization was below 99% on three of the trailing seven days. Consecutive firing days with gaps of a week or less form one episode. Outcomes are the verified outages of Table 2 and removals from the top tier, with outcomes of one organization closer than a week merged; there are 16. An episode is a *warning* if an outcome of its organization follows between one week and the window later, an *incident* if an outcome falls within a day before to a week after its start (the episode is the outage itself, or too late to help), and a *false alarm* otherwise; precision counts warnings against false alarms, and lead time runs from the earliest warning. Validators are dropped from the computation after their last day with any validating time, so that retired validators still listed in quorum sets do not read as permanent failures.

Rule	Window	Warnings	Incidents	False alarms	Precision	Outcomes warned	Recall	Lead, median (range)
S3	180 days	16	4	12	57%	9 of 16	56%	153 (15 to 178)
S3	365 days	18	3	11	62%	10 of 16	62%	174 (15 to 347)
S2	365 days	13	0	2	87%	7 of 16	44%	174 (22 to 334)

Table 3: Early warning rules scored over all top tier organizations, 2019 to 2026.

Under rule S3 the lead times of the cases above are 155 days for Wirex (from 11 July 2022 to 13 December, its last full day in the top tier), 153 days for Keybase and 178 days for SatoshiPay’s August 2025 outage. The figures are dominated by one organization: 12 of S3’s 18 warnings and 12 of S2’s 13 are SatoshiPay, whose chronic degradation was followed by one of its eight outcomes within a year almost every time. Without SatoshiPay, S3 has 6 warnings against 11 false alarms, a precision of 35%, and warns of 5 of the 9 remaining outcomes. Franklin Templeton and LOBSTR account for seven of the eleven false alarms. A looser rule (any validator below 90% or the organization below 99%) flags Wirex 227 days ahead, but only 44% of its 109 episodes are warnings.

The conclusion is the opposite of the one the three featured cases suggest on their own: availability is a reasonable screen (it would have flagged three of the four organizations that eventually left, months ahead, and it tracks a chronically unstable organization well) but a poor alarm for the others, because most of their degradation is transient. Whether combining it with the margin itself, so that a degrading organization matters more when the margin is already thin, yields a usable signal is an open question, and the proposed work treats it as one.

5. Current state

As of 1 October 2026 the top tier has 30 validators in ten organizations (Blockdaemon, Creit Technologies, Figure, Franklin Templeton, LOBSTR, MoneyGram, OBSRVR, Public Node, Range, SDF), threshold seven of ten, a nominal and effective liveness margin of four organizations, and a top tier splitting set of four. A drop of the organization level splitting set to one and briefly to zero, intermittently between 08:09 and 22:45

UTC on 1 October 2026 and gone by 23:00, with no quorum set change and no validator failure, coincides with a change in Radar’s analysis code that computes the splitting set network wide [9] and is treated as a measurement artifact.

6. Limitations

- **Single vantage point.** A validator recorded as not validating or unreachable may have been invisible to Radar’s crawler only. The one event with a public record disagrees with Radar on timing: SDF’s statement puts the start of its halt on 6 April 2021 at about 08:00 UTC, while Radar still recorded SDF’s validators as validating at 09:58 UTC and saw them down from 14:34 UTC. The discrepancy is unresolved.
- **Nominal is not configured.** Nominal margins react to reachability as well as to configuration, and a failed validator that is also unreachable lowers the nominal margin rather than the effective one, so a degradation event can start hours after the failure (Wirex on 13 July 2022). Only snapshot diffs establish a configuration change, and we classify only on that basis.
- **Sampled membership and verification.** Membership is read from 52 snapshots, one at the start of each stable configuration state, and validator status from 159 snapshots in all, not from every scan; an organization that joined and left between samples would be missed, each event is verified at one to four instants rather than reconstructed in full, and twelve short dips could not be attributed at the sampled instant.
- **Operator intent is unknown.** The four configuration changes may have been deliberate temporary measures. We show their effect on the network, not that they were mistakes.
- **Scoring thresholds.** The 10%, 90%, 99% and seven day parameters of the rules are starting points chosen by hand on the full record, and the scores are dominated by one organization; the proposed work will choose thresholds on 2019 to 2023 and test on 2024 to 2026.
- **Top tier only.** The blocking set recomputation considers the transitive quorum set and ignores validators outside it.
- **Analysis backend.** Radar’s analyzer changed over the period (its own code, then `fbas_analyzer`, now `python-fbas`). Our independent recomputation covers the four configuration events; the degradation series still relies on Radar’s figures.
- **No per event public record.** Apart from 6 April 2021 we found no public report for any of these events, so the data are the only evidence.

7. Implications for the proposed work

The findings point to two concrete pieces of work, both narrower than the original proposal, and one measured question.

1. **A complete longitudinal study.** Classify every quorum set change in the seven year record by its effect on network wide margins, separating configuration changes from reachability effects, reconstruct each degradation period in full, and release the dataset and scripts.
2. **A check before the change.** Given a proposed quorum set and the current network state, report the network wide minimal blocking and splitting sets that would result, built on `python-fbas` [8]. The four cases above are its first test set; the expansion of the tier 1 set toward thirteen organizations, in which each operator updates its own quorum set independently [11], is its first live use. The same analysis can evaluate alternatives to the current threshold rule and connect to the theory of reconfigurable heterogeneous quorum systems [12].
3. **An early warning signal at organization level,** scored for lead time, precision and recall as in Table 3, with the margin itself as an input, and reported whether or not it turns out to be useful.

References

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